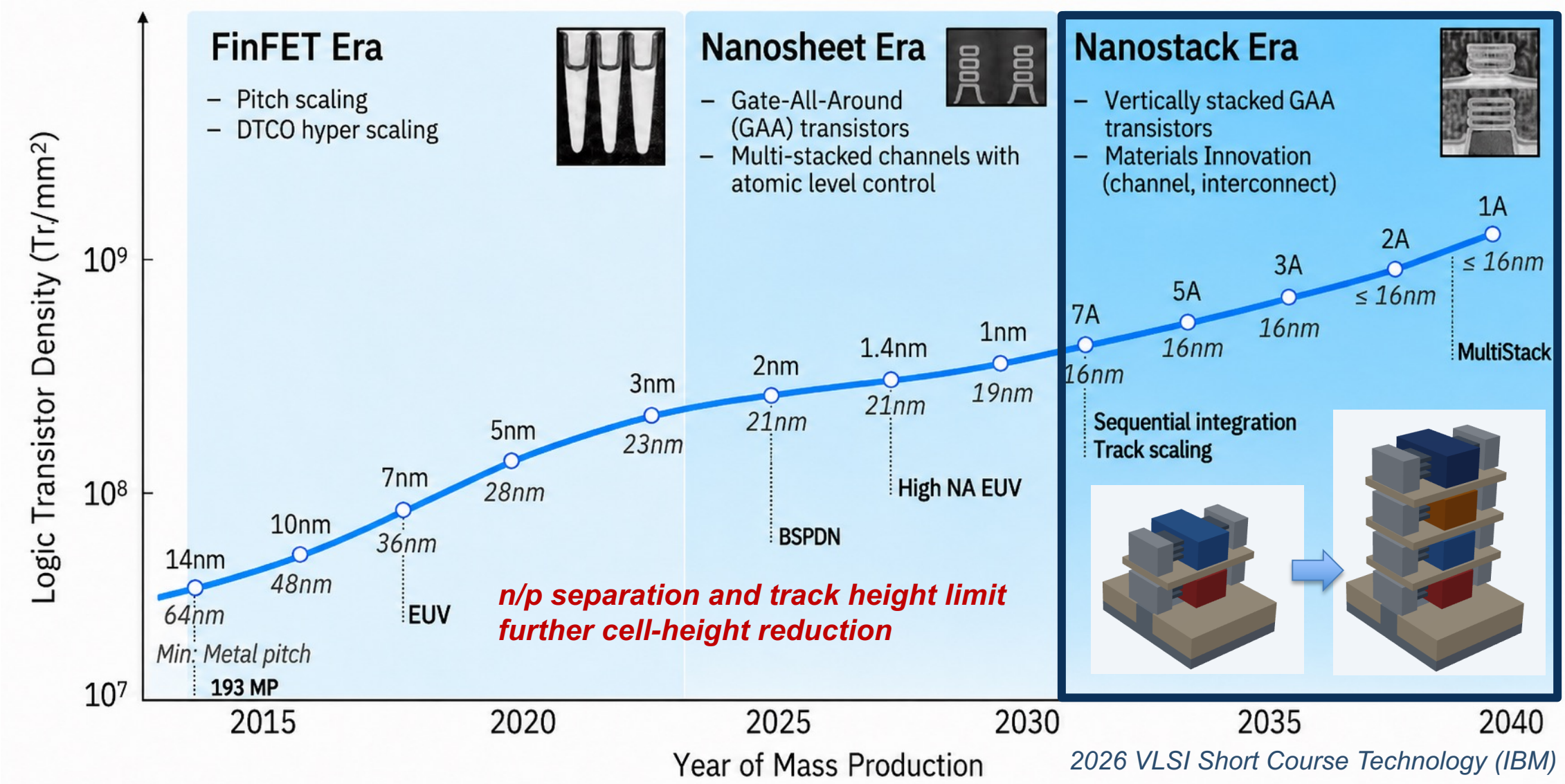

FinFET Era
– Pitch scaling
– DTCO hyper scaling
Nanosheet Era
– Gate-All-Around (GAA) transistors
– Multi-stacked channels with atomic level control
Nanostack Era
– Vertically stacked GAA transistors
– Materials Innovation (channel, interconnect)
Logic Transistor Density (Tr./mm²)
10⁹
10⁸
10⁷
14nm
64nm
Min: Metal pitch
193 MP
10nm
48nm
7nm
36nm
EUV
5nm
28nm
3nm
23nm
2nm
21nm
BSPDN
1.4nm
21nm
High NA EUV
1nm
19nm
7A
16nm
Sequential integration
Track scaling
5A
16nm
3A
16nm
2A
≤ 16nm
1A
≤ 16nm
MultiStack
n/p separation and track height limit further cell-height reduction
2015
2020
2025
2030
2035
2040
Year of Mass Production